

Rajalakshmi Engineering College

Name: Amirita varshini M
Email: 241001014@rajalakshmi.edu.in
Roll no: 241001014
Phone: 9344660582
Branch: REC
Department: IT - Section 1
Batch: 2028
Degree: B.E - IT

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Monica is interested in finding a treasure but the key to opening is to get the sum of the main diagonal elements and secondary diagonal elements.

Write a program to help Monica find the diagonal sum of a square 2D array.

Note: The main diagonal of the array consists of the elements traversing from the top-left corner to the bottom-right corner. The secondary diagonal includes elements from the top-right corner to the bottom-left corner.

Input Format

The first line of input consists of an integer N, representing the number of rows and columns.

The following N lines consist of N space-separated integers, representing the 2D array elements.

Output Format

The first line of output prints "Sum of the main diagonal: " followed by an integer, representing the sum of the main diagonal.

The second line prints "Sum of the secondary diagonal: " followed by an integer, representing the sum of the secondary diagonal.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

1 2 3

4 5 6

7 8 9

Output: Sum of the main diagonal: 15

Sum of the secondary diagonal: 15

Answer

// You are using Java

```
import java.util.*;
```

```
class main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int a=sc.nextInt();
```

```
        int b[][]=new int[a][a];
```

```
        for(int i=0;i<a;i++){
```

```
            for(int j=0;j<a;j++){
```

```
                b[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        int f=0;
```

```
        int g=0;
```

```
        for(int i=0;i<a;i++){
```

```
            f+=b[i][i];
```

```
            g+=b[i][a-i-1];
```

```
        }
```

```
        System.out.println("Sum of the main diagonal:"+f);  
        System.out.println("Sum of the secondary diagonal:"+g);  
    }  
}
```

Status : Correct

Marks : 10/10